

Introduction to Quadratic Relationships and Modeling

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Quadratic Relationship

A **quadratic relationship** is a relationship that can be represented by a **quadratic equation** or **quadratic function** of the form

$$y = ax^2 + bx + c,$$

where $a \neq 0$.

Independent and Dependent Variables

The **independent variable** is the quantity you choose or control.

The **dependent variable** changes based on the independent variable.

Read the Problem

Read the problem carefully to identify the **known quantities** and the **unknown quantity**.

Define a Variable

Define a variable to represent the **unknown quantity**.

Write an Equation

Write an equation that represents the situation.

Standard Quadratic Form

Simplify the equation until it is in **standard quadratic form** whenever possible.

$$ax^2 + bx + c = 0$$

Examples

Let's model real-life situations using quadratic equations.

Example 1

The length of a rectangle is 5 m longer than its width. Its area is 84 m².

Let the width be x .

$$\text{Length} = x + 5$$

$$x(x + 5) = 84$$

$$x^2 + 5x - 84 = 0$$

$$\boxed{x^2 + 5x - 84 = 0}$$

Example 2

A square garden has an area of 196 m^2 . Let its side length be x .

$$x^2 = 196$$

$$x^2 - 196 = 0$$

Example 3

A ball is thrown upward. Its height after t seconds is

$$h = -5t^2 + 20t + 3.$$

The equation contains the squared variable t^2 .
Therefore, it is a **quadratic relationship**.

Example 4

A farmer uses 40 m of fencing for the other three sides of a rectangular pen beside a wall.
Let x be the width.

Two widths use $2x$ m of fencing.

$$\text{Length} = 40 - 2x$$

Area:

$$A = x(40 - 2x)$$

$$\boxed{A = -2x^2 + 40x}$$

Example 5

Ticket sales are modeled by

$$R = -2n^2 + 120n.$$

The equation contains the squared variable n^2 .
Therefore, it is a **quadratic relationship**.

Example 6

The base of a triangle is 3 cm more than its height. Its area is 54 cm^2 .
Let the height be x .

$$\text{Base} = x + 3$$

Using

$$A = \frac{1}{2}bh,$$

$$\frac{1}{2}(x + 3)x = 54$$

$$x(x + 3) = 108$$

$$\boxed{x^2 + 3x - 108 = 0}$$

Practice 1

The length of a rectangle is 5 m longer than its width. Its area is 84 m^2 .

Represent the situation using a quadratic equation.

Practice 2

A square garden has an area of 196 m^2 . If its side length is represented by x , write the quadratic equation that models the situation.

Practice 3

A ball is thrown upward. Its height after t seconds is given by

$$h = -5t^2 + 20t + 3.$$

Identify why this situation is modeled by a quadratic relationship.

Practice 4

A farmer plans to build a rectangular pen beside a wall using 40 m of fencing for the other three sides.

Let x be the width. Write the quadratic expression for the area.

Practice 5

A school's ticket sales are modeled by

$$R = -2n^2 + 120n,$$

where n is the number of tickets sold in hundreds.

Explain why this is a quadratic relationship.

Practice 6

The base of a triangle is 3 cm more than its height. If its area is 54 cm^2 , represent the situation using a quadratic equation.

Activity 1

The product of two consecutive positive integers is 342.

Represent the situation using a quadratic equation.

Activity 2

A stone is thrown upward. Its height after t seconds is given by

$$h = -5t^2 + 30t + 2.$$

Identify why this situation is modeled by a quadratic relationship.

Activity 3

The product of two consecutive even integers is 440.

Represent the situation using a quadratic equation.

Activity 4

The length of a rectangle is 6 cm greater than its width. If its area is 160 cm^2 , write the quadratic equation representing the situation.

Activity 5

The base of a triangle is 5 cm more than its height. If its area is 75 cm^2 , represent the situation using a quadratic equation.

Thank you!